

DATA ANALYSIS PROJECT REPORT

Global University Performance and Habit

Project Title	Global University Performance and Habit Data analysis
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Course / Section	01412[section D]
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1. Executive Summary

This project analyzed a student dataset containing academic, demographic, and lifestyle-related variables such as GPA, study hours, class attendance, sleep duration, screen time, mental stress level, and final exam scores. The main goal was to identify factors influencing student performance and to evaluate academic efficiency using both statistical analysis and custom metrics. The analysis was performed using Pandas for data cleaning, grouping, and aggregation; NumPy for creating custom calculations such as academic efficiency and healthy score; and Matplotlib/Seaborn for visualization including bar charts, pie charts, histograms, and correlation heatmaps. The most important findings showed that higher study hours and better attendance are strongly associated with higher GPA and exam scores. Additionally, high screen time and stress levels were linked to lower academic efficiency, while country-wise analysis revealed differences in the distribution of high and low-performing students. Overall, the results suggest that both academic habits and lifestyle factors significantly influence student performance.

2. Introduction and Project Objectives

Item	Your Content
Dataset name	Global University Performance and Habit
Dataset source	Kaggle
Main project goal	The goal of this project is to analyze how student lifestyle, study habits, attendance, screen time, sleep patterns, and mental stress affect academic performance.
Research Question 1	We are analyzing which countries have the highest and lowest number of academically efficient students.
Research Question 2	We will investigate what factors contribute to high and low academic efficiency among students.
Research Question 3 (optional)	We will analyze how does mental stress level affect student's final exam scores.

3. Dataset Description

This dataset contains information about students' academic performance, lifestyle habits, and personal factors. Each row in the dataset represents one individual student. The dataset includes variables related to demographics, study habits, attendance, GPA, sleep hours, screen time, gaming habits, mental stress level, exercise, part-time jobs, and final exam scores, ai tool usage hours, extracurricular hours per week.

The dataset also includes academic-related variables such as study hours per day, class attendance percentage, GPA, exam preparation days, and final exam scores. Lifestyle-related variables include sleep hours, total screen time, gaming hours, extracurricular activities, and mental stress level.

3.1 Basic Dataset Information

Property	Value
Number of rows	10000
Number of columns	27
Data types present	Numerical and Categorical mixed
Target or key variable(s)	'student_id', 'age', 'gender', 'country', 'gpa', 'study_hours_per_day', 'class_attendance_percent', 'sleep_hours', 'mental_stress_level', 'final_exam_score', 'total_screen_time', healthy_score
Potential issues observed at first glance	Missing value

3.2 Initial Observations

Missing values were identified in the favorite_AI_tool column using the `isnull().sum` function. This analysis helps detect incomplete records and ensures better data quality before performing further analysis.

4. Data Cleaning and Preparation

Issue Found	Action Taken	Reason / Justification	Code Reference
Missing value in favourite_ai_tools column	Using <code>with dropna(subset=...)</code> and also use <code>bfill</code> , <code>fill</code> to fill the box	It was used to remove rows containing missing values in specific columns. This helps ensure accurate analysis by eliminating incomplete data records.	7-18
Invalid values in gpa column	Using <code>quantile()</code> and IQR method	The IQR method was chosen because it is simple, reliable, and effective for detecting extreme values without being heavily affected by the data distribution.	46-51
Invalid values in study per day hour	Using <code>quantile()</code> and IQR method	Extremely low or high study-hour values were treated as unusual observations because they may negatively affect the analysis results.	46-51
Unusual values in screen time hour	Using <code>quantile()</code> and IQR method	To remove outliers invalid value and look data set realistic	46-51

5. Feature Engineering

New Feature	How It Was Created	Why It Is Useful
Total screen time	<code>df['total_screen_time'] = (df['screen_time_hours'] + df['gaming_hours'])</code>	combine students overall daily screen usage into a single variable. It is help to academic efficiency found.
Healthy score	<code>df['healthy_score'] = (df['exercise_hours_per_week'] + df['sleep_hours'])</code>	The healthy_score was created to combine exercise and sleep into a single health-related metric. It helps analyze how healthy lifestyle habits may influence academic performance, and get final score.
academic efficiency	<code>df['academic_efficiency'] = (df['gpa'] * df['class_attendance_percent']) / (df['study_hours_per_day'] + 1)</code>	To found which country student have low and high efficiency student

6. Analysis Methodology

1. Pandas was used for data cleaning, handling missing values, grouping data, filtering records, and performing comparative analysis on student performance and efficiency.
2. NumPy was used to create custom metrics such as academic efficiency , as well as for numerical calculations and condition-based analysis.
3. Matplotlib and Seaborn were used to create visualizations including bar charts, pie charts, histograms, and heatmaps to identify patterns, comparisons, and relationships in the dataset.
4. Correlation analysis, subgroup comparison, outlier detection, and distribution analysis were performed to answer the project research questions and identify meaningful insights from the student dataset.

7. Analysis and Visualizations

7.1 Research Question 1

Which countries have the highest and lowest number of academically efficient students?

Academic efficiency scores were calculated for all students and divided into high-efficiency and low-efficiency groups using percentile-based filtering. The number of high and low efficiency students was then grouped by

country using `value_counts()`, and the results were visualized using pie charts to compare country-wise student efficiency distributions.

Figure1:

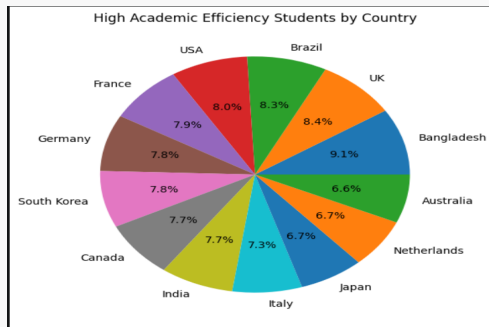


Figure 2:

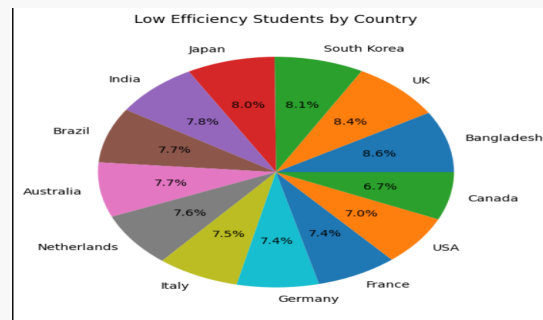


Figure1: High Academic Efficiency Student by country Figure2: High Academic Efficiency Student by country

The percentage distribution of high academic efficiency students across different nations. Bangladesh has the largest share at 9.1%, while Australia holds the lowest share at 6.6%. Bangladesh has the highest share of low-efficiency students at 8.6%, followed closely by the UK at 8.4%. Canada accounts for the lowest share in the chart at 6.7%, just below the USA at 7.0%

7.2 Research Question 2

What factors contribute to high and low academic efficiency among students?

Grouped comparisons were then performed on variables such as GPA, study hours per day, class attendance percentage, sleep hours, total screen time, mental stress level and healthy score and identify what is the reason behind academic efficiency. Academic efficiency is divided into 3 categories by low, medium and high

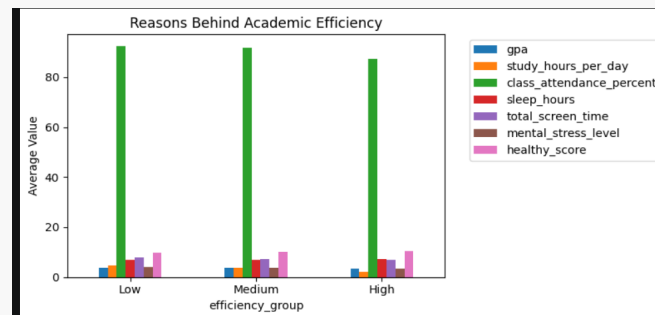


Figure 3: Reasons Behind Academic Efficiency

Moderate stress and balanced screen time characterize the medium academic efficiency group. Increased daily screen time correlates directly with reduced academic output, showing its highest values among low-performing individuals.

7.3 Research Question 3

How do exam preparation days influence students' average final exam scores?

Exam preparation days and the average final exam score for each stress category was calculated using `groupby()` and mean aggregation.

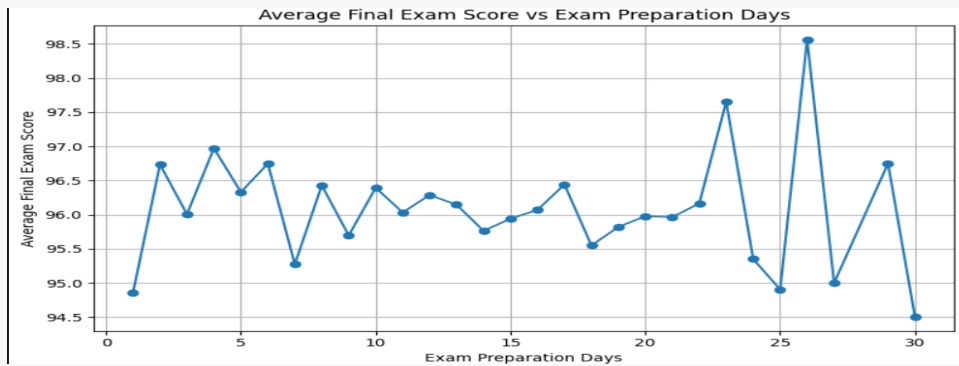


Figure 4: Average final exam score vs exam preparation days

Short-Term Prep (0–5 Days): Scores start high at roughly 97 .

Mid-Term Prep (6–20 Days): The line stabilizes into a tighter, lower range, mostly fluctuating between 95.5 and 97.0. This indicates a period of consistent but slightly lower average outcomes.

Long-Term Prep (21–30 Days): This zone shows the most extreme volatility. It features a dramatic rise to the overall highest peak of 98.5 at 26 days, followed immediately by a sharp plunge to the graph's lowest trough near 94.6 at 29 days.

8. NumPy-Based Custom Computation

[Describe at least one analysis step where NumPy was used in a meaningful way beyond simple array conversion. Examples include z-score computation, normalization, percentiles, `np.where`-based classification, binning, or a custom score.]

Component	Description
Computation used	Academic Efficiency score calculation
Purpose	To measure how effectively students achieve academic performance relative to their study hours and attendance
Formula or logic	$\frac{(\text{GPA} \times \text{Class Attendance Percentage})}{(\text{Study Hours Per Day} + 1)}$
Result / takeaway	Students with higher GPA and attendance generally achieved higher academic efficiency, while students with lower academic performance or inefficient study habits showed lower efficiency scores.

9. Key Findings

1. Students who studied more hours per day generally achieved higher GPA and final exam scores, showing a strong positive relationship between study habits and academic performance.
2. High academic efficiency students typically had higher attendance percentages and lower screen time compared to low efficiency students.
3. Higher mental stress levels were associated with increased total screen time, suggesting that excessive screen usage may contribute to student stress.
4. Some students with very high study hours still showed low academic efficiency, indicating that study quality and lifestyle factors are also important for academic success.
5. The country-wise analysis showed that some countries had a higher proportion of high academic efficiency students, indicating stronger academic performance and balanced study habits within those groups.

10. Limitations

The limitation of this project was the presence of missing and invalid values in the favorite_AI_tool column. Some records contained unclear or inconsistent entries, which made the analysis of preferences more difficult. These issues may have slightly affected the accuracy of results related to usage patterns.

The analysis suggests that students with higher attendance and lower screen time tend to achieve better academic efficiency. However, the project cannot prove that screen time or stress directly causes poor academic performance because other external factors may also influence student outcomes.

11. Conclusion

The objective of this project was to analyze student academic performance and identify the factors influencing high and low academic efficiency. Using Pandas, NumPy, Matplotlib, and Seaborn, multiple analyses were performed on study habits, attendance, sleep, screen time, stress levels, and exam performance.

The analysis showed that students with higher study hours and better attendance generally achieved better GPA and final exam scores. High academic efficiency students were also associated with healthier lifestyle patterns such as lower screen time and balanced stress levels. In contrast, low efficiency students often showed higher stress and screen usage. Country-based comparisons also revealed differences in the distribution of high and low efficiency students.

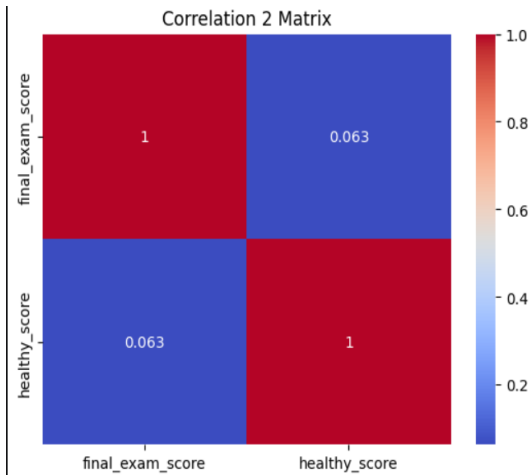
Overall, the project suggests that academic success is influenced not only by study time but also by lifestyle and behavioral factors. Future analysis could include larger datasets, additional psychological or environmental variables, and predictive modeling techniques to gain deeper insights into student academic performance.

12. References / Dataset Source

global_university_students_performance_habits_10000 (1).csv – Kaggle

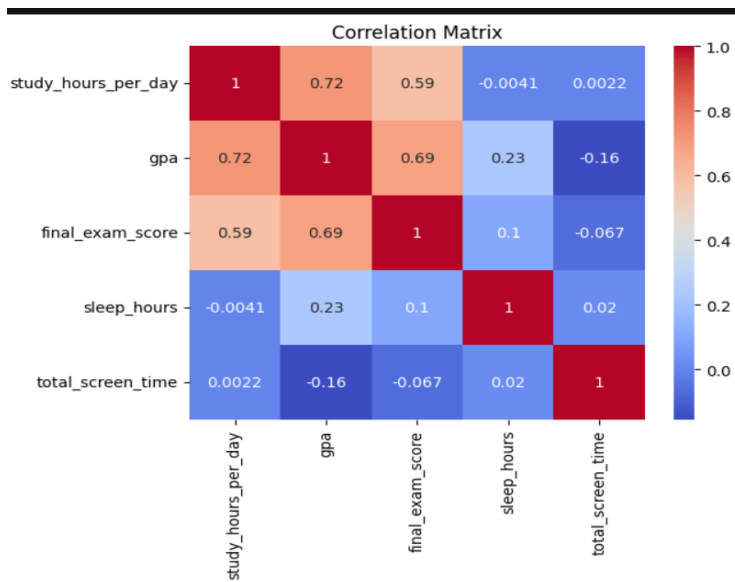
13. Appendix (Optional)

Correlation about healthy score vs final score



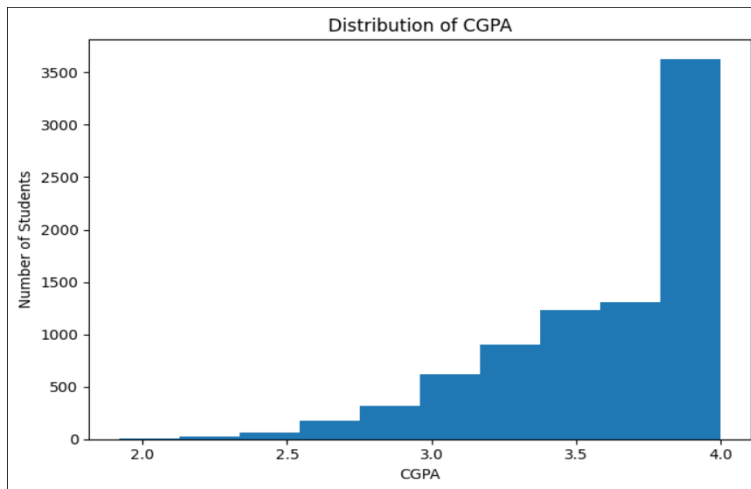
A correlation of 0.063 is very close to 0

That means there is almost no linear relationship between `healthy_score` and `final_exam_score`



A correlation matrix is created by:

- Taking multiple numerical variables (like study hours, GPA, exam score, etc.)
- Comparing every variable with every other variable
- Calculating a correlation coefficient (Pearson correlation) for each pair



Most students are concentrated at the very top, with a CGPA between 3.5 and 4.0. The graph is heavily left-skewed, showing that almost no students have low grades. This high concentration suggests either a very selective student group or lenient grading.

Final Submission Checklist

- ✓ I included my GitHub repository link or Google Colab notebook link in the Google Sheet provided by the instructor.
- ✓ I attached the project report as a separate file.
- ✓ My report includes dataset source, project objective, cleaning steps, feature engineering, analysis, visualizations, findings, limitations, and conclusions.
- ✓ My code uses Pandas, NumPy, and Matplotlib meaningfully.
- ✓ All bracketed placeholder text and helper notes were replaced or deleted before submission.